



FOOD FOR THOUGHT

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IS IT SAFE FOR ATHLETES TO WORK OUT IN FASTED STATE?

Fasting before exercising, especially for athletes, can be a **tricky matter**. It largely depends on the individual, the type of exercise, and the duration and intensity of the workout.

Have you ever wondered if it's safe for you to exercise when you haven't eaten anything? Think of your body as a car, and your workout is like a trip you're going on with your car. Eating before a workout is like putting fuel in your car before the trip. Just like how you wouldn't want your car to run out of gas while you're driving, you also don't want your body to run out of energy during your workout. However, when you fast, fasting before a workout is like letting your body use up its stored fuel before giving it more.

Fasting before exercising, especially for athletes, can be a tricky matter. It largely depends on the individual, the type of exercise, and the duration and intensity of the workout.

During low-intensity exercise, such as a light jog or steady-state cycling, the body primarily relies on fats as a fuel source which means your body mainly uses fats to give you energy during this type of exercise. Fats provide a steady and abundant source of energy, making them ideal for prolonged activities. In contrast, during high-intensity exercise, like sprinting or intense weightlifting, and longer duration the body shifts its reliance on carbohydrates (glycogen) for energy.

Now in the case of recreational athletes, who play a sport just for the joy of it, when they train in a fasted state, fat oxidation is the primary energy pathway. Exercising in a fasted state may not significantly impact performance. However, the story changes during high-intensity exercises for longer duration, typically how most of the athletes train. Without readily available carbohydrates, elite athletes may experience a decrease in performance due to limited glycogen stores and impaired glycolytic metabolism. The absence of carbohydrates can lead to decreased muscle glycogen levels, resulting in fatigue, reduced power output, and compromised exercise capacity.

Despite the potential drawbacks, some elite athletes choose to incorporate fasted training into their regimen for specific purposes. One purported benefit is the enhancement of fat oxidation and metabolic flexibility. By training in a fasted state, athletes aim to increase their ability to use fats as a primary fuel source, which may be advantageous for endurance events or activities with prolonged duration.

Moreover, fasting has been associated with various metabolic adaptations, including improvements in insulin sensitivity, mitochondrial function, and cellular

repair processes. These adaptations could potentially enhance athletic performance and recovery. Therefore to optimise performance, incorporate fasting twice a week, strategically timed with low-intensity training sessions, while eating a pre-workout meal on the remaining days before training.

Engaging in low-intensity exercise during fasting periods can help preserve muscle glycogen stores while promoting fat oxidation, thereby mitigating the risk of performance decrements associated with high-intensity efforts in a fasted state. Fasting also triggers a process known as mitochondrial biogenesis. Imagine your body is like a big city. Every city needs power to run, right? Well, inside your body, there are these tiny, tiny power plants called mitochondria. They make energy for your body to work, just like how power plants make electricity for a city. Now, mitochondrial biogenesis is like building more power plants in the city. When your body needs more energy, it makes more mitochondria. It's like adding more power plants to make sure the city has enough electricity to keep everything running smoothly.

Just like that, mitochondrial biogenesis gives you more energy, enhancing your performance.

On days designated for pre-feed